25901913 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMAT202 - ESSENTIALS OF APPLIED MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

{Short Answer Questions}

Answer any **5** questions

Each question carries **2** marks

1. Prove that $\lim_{z\to\infty}\left[\frac{2z+25}{3z-37}\right]=\frac{2}{3}$ [U] / [CO1]
2. Check whether $u(x,y)=x^2+y^2$ is harmonic. [U] / [CO1]
3. Find the integrating factor for the differential equation $y'+y=\frac{1}{1+e^{2x}}$. [U] / [CO2]
4. Solve the differential equation $\frac{dy}{dx}=\frac{y}{x}$. [U] / [CO2]
5. Find any two polar coordinates of the point $(3,\frac{\pi}{4})$. [K] / [CO3]
6. Define an ellipse. Also define foci of the ellipse. [K] / [CO3]
7. Define the unit tangent vector to a smooth curve $\vec{r}(t)$. [K] / [CO4]
8. Find $\lim_{t\to 0}\left[\cos t\,\hat{i}+\sin t\,\hat{j}-t\,\hat{k}\right]$. [U] / [CO4]

[2x5 = 10]

Part B

{Short Essay Questions}

Answer any **5** questions

Each question carries **6** marks

9. Verify that $z = 1 - i$ satisfies the equation $z^2 - 2z + 2 = 0$. [U] / [CO1]

10. Determine the real and imaginary parts of $f(z) = \dfrac{1}{1-z}$. [U] / [CO1]

11. Given $z_1 = 2e^{i\pi/3}$ and $z_2 = 3e^{i\pi/6}$, find the value of the product $z_1 z_2$ in the form $a + ib$. [U] / [CO1]

12. Find the particular solution for the differential equation $\dfrac{dy}{dx} + y\tan x = \sin x$ that satisfies the initial condition $y = 1$ when $x = \dfrac{\pi}{4}$. [An] / [CO2]

13. Find the vertex, focus, directrix and equation of the axis of the parabola $x^2 = 6y$. Also, sketch the parabola. [U] / [CO3]

14. Find the vertices, foci, eccentricity and directrices of the hyperbola $5x^2 - 4y^2 = 20$. [U] / [CO3]

15. Find the curvature of the curve $\vec{r}(t) = t^2\,\hat{i} + t^2\,\hat{j} + t^2\,\hat{k}$. [U] / [CO4]

16. The position vector of a particle at time t is given by $\vec{r}(t) = t^2\,\hat{i} + (2t-1)\,\hat{j} + t^2\,\hat{k}$. Find the particle's velocity, speed, direction of motion and acceleration at $t = 2$. [A] / [CO4]

[6x5 = 30]

Part C

{Essay Questions}

Answer any **3** questions

Each question carries **10** marks

17. Consider the function $f(z) = e^{-z}$. [U] / [CO1]
    (a) Express $f(z)$ in the form $u(x,y) + iv(x,y)$.
    (a) Using Cauchy-Reimann equations, show that $f(z)$ is analytic everywhere in the complex plane.
    (c) Find $f'(z)$ and $f'(0)$.

18. Show that the differential equation $(1 + 4x\,y + 2y^2)dx + (1 + 4x\,y + 2x^2)dy = 0$ is exact and solve it. [U] / [CO2]

19. (a) Check whether the function $y = e^x$ is a solution of the differential equation $y'' - 5y' + 4y = 0$. [U] / [CO2]
    (b) Verify that the function $y = C_1 e^{4x} + C_2 e^{-2x}$ is a solution of the differential equation $y'' - 2y' - 8y = 0$.

20. Identify the following conics and find their directrix that corresponds to the focus at the origin: [A] / [CO3]

(a) $r = \frac{12}{3+\ 3\ cos\theta}$

(b) $r = \frac{4}{2-sin\theta}$.

21. (a) Find ∇f for the function $f(x, y, z) = e^x + y\ \cos z + (y + 1) \sin x$ at the point $(0, 0, 0)$. [U] / [CO4]

(b) Find the directions in which the function f$(x, y) = x^2 + x\,y + y^2$, increases most rapidly and decreases most rapidly at $(-1, 1)$.

22. Find the length of the indicated portions of the following curves: [U] / [CO4]

(a) $\vec{r}(t) = (2 + 5t)\ \hat{i} - (2t + 1)\ \hat{j} + t\ \hat{k}, \quad 0 \le t \le 3$

(b) $\vec{r}(t) = 6 \sin 2t\ \hat{i} + 6 \cos 2t\ \hat{j} + 5\,t\ \hat{k}, \quad 0 \le t \le \pi.$

[10x3 = 30]